

ARYAN SAINI

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RESEARCH VISION AND INTERESTS

My vision is to develop interactive systems that empower individuals to harness their body's full potential. I aim to design wearable technological augmentations that seamlessly integrate into daily life, enhancing physical capabilities and facilitating a deeper connection between the body and technology. These systems will not only augment physical abilities but also provide intuitive and personalized experiences, fostering a symbiotic relationship between humans and technology.

Human-Computer Interaction (Bodily Augmentations, Embodied Interaction, Soft Exoskeletons, Pneumatic Interfaces, Virtual Reality, and Haptics)

EDUCATION

Exertion Games Lab, Monash University

February 2022 - August 2025 (Projected)

Ph.D. in Human-Computer Interaction

Advisors: [Prof. Floyd Mueller](#) and [Prof. Elise van den Hoven](#)

Thesis: **Designing Pneumatic Bodily Extensions for Promoting Embodiment across Everyday Life Experiences**

IIIT-Delhi, New Delhi, India

August 2015 - May 2019

Bachelor of Technology (Electronics and Communications Engineering)

Undergraduate Thesis: **Designing Wearable Trinkets and Toolkits**

RESEARCH EXPERIENCE

Weave Lab, IIIT-Delhi, New Delhi

Jan 2020 - May 2021

Research Assistant

*Advisor: **Prof. Aman Parnami***

Microsoft Research, India

June 2019 - Dec 2019

Research Intern

*Advisor: **Dr. Manohar Swaminathan***

SELECTED PUBLICATIONS

- [1] **Saini, A.**, Sridhar, S., ..., van den Hoven, E., Mueller, F. F. Pneunocchio: Understanding the Design of a Nose-based Bodily Extension that Suggests Lying. **CHI 2025 (Under review)**. [\[PDF\]](#) [\[Video\]](#)
- [2] **Saini, A.**, Sridhar, S., ..., van den Hoven, E., Mueller, F. F. 2024. PneuMa: Designing Pneumatic Bodily Extensions for Supporting Movement in Everyday Life. **TEI 2024**. [\[PDF\]](#) [\[Video\]](#)
- [3] **Saini, A.**, Bhatia, A., Kalra, I., Mukherjee, M., Parnami, A. 2023. DUMask: A Discrete and Unobtrusive Mask-Based Interface for Facial Gestures. **AHs 2023** [PDF](#)
- [4] Arora, J., **Saini, A.**, Mehra, N., Jain, V., Shrey, S., Parnami, A. 2019. VirtualBricks: Exploring a Scalable, Modular toolkit for Enabling Physical Manipulation in VR. **CHI 2019**. [PDF](#)

TECHNICAL SKILLS

**Programming Languages
Tools**

C, Embedded C, Python, JavaScript and Verilog
OnShape, Blender, AutoCAD, Git, MATLAB, Wireshark,
L^AT_EX, Xilinx, LTSpice

**Hardware Skills
Multimedia and Design**

3D Design and Printing, PCB Fabrication, Communication Protocols
Adobe Premiere Pro, Photoshop, Illustrator, After Effects, Sketch,
Audacity and Fritzing

RELEVANT COURSEWORK

Wearable Applications, Research, Devices, and Interactions (WARDI), Robotics, Embedded Logic Design, Computer Architecture, Computer Networks, Radar Systems